
EP 401P: Engineering of Instrumentation

Course Code	EP 401P
Course Name	Engineering of Instrumentation
L-T-P-C	1-0-5-4
Prerequisites	PH301 or PH513 and PH501 or PH523
Intended for	B.Tech.
Distribution	Core
Approval	20th Senate (Apr 2019), in the B.Tech Engineering Physics curriculum; 37th BoA (Oct 2020); 30th Senate (Feb 2021), in the B.Tech Engineering Physics curriculum

Course Contents

- **Module I**

Introduction to data Acquisition systems, Labview/open source (such as Python) programming (Interfacing and programming), Signal processing and Error analysis. [Lectures]

- **Module II**

Automatic control (PID control, Feed forward control, Time delay and inverse response systems, Sequence control). [Lectures]

- **Module III**

Cryogenics Instrumentation (Low temperature, Liquefaction of gases, Close Cycle Refrigerator, Temperature sensor). Vacuum pumps (Rotary, Dry scroll, Root pumps) with focus on role of valves, gauges etc. [Lectures]

- **Module IV**

Introduction to design and working of instruments (Electron microscope, Scanning Tunneling Microscopy, Atomic force microscope and Superconducting magnets) [Lectures]

Text Books

1. C. Barry Carter and David B. Williams, **Transmission Electron Microscopy**, Springer, New York (2016).
2. Robert C. Richardson and N. Smith, **Experimental Techniques in Condensed Matter Physics at Low Temperatures**, CRC Press (2018).